

ADRIANA DELGADO LEE

Burnaby, BC
<https://adri2765.github.io/>

adri.del.lee@outlook.com
[linkedin.com/in/adridelLee/](https://www.linkedin.com/in/adridelLee/)

SKILLS

Hardware

3D Printing
Laser Cutting
Mechanical / Electrical / Software
Prototyping
Breadboarding
Soldering (Manual & Hot Air)
Machine Tools

Software

SolidWorks & SW PDM
AutoCAD
Microsoft Office
Adobe Illustrator & Photoshop
Bluebeam Revu
Revizto
Revit

Coding

C++
Java Script
Java
Python
MATLAB
HTML
VS Code

Languages

English - Professionally Fluent
Spanish - Fluent

WORK EXPERIENCE

PowerTech Labs - BC Hydro

Surrey, BC

Manufacturing Co op Student

Sept. 2024 - Dec. 2024

CADD Technician

Jan. 2025 - Aug. 2025

- Reviewed and updated mechanical drawing packages for hydrogen fueling stations, applying GD&T principles to ensure precision, manufacturability, and compliance with safety standards while collaborating with manufacturing to resolve design challenges.
- Designed, modelled, and prototyped hydrogen testing systems, including a high-pressure automated test bench and a hydrogen particulate test jig, supporting full lifecycle development from P&ID creation to fabrication and hands-on assembly.
- Redesigned critical station subsystems (instrument air and compressed gas storage panels), improving safety, reducing footprint, and enhancing cost efficiency.
- Built and standardized a SolidWorks and AutoCAD component library of 500+ parts to accelerate design and ensure model consistency across projects.
- Modelled complex hydrogen tubing systems and structural supports using SolidWorks Weldments and Design Tables, enabling fabrication of assemblies comprising hundreds of high-pressure components.

Smith + Andersen

Burnaby, BC

Jr. Mechanical Designer

Jan. 2023 - Sept. 2023

- Designed and reviewed mechanical, HVAC, fire protection, water/IV, and drainage systems for the New St. Paul's Hospital Project using Revit, Bluebeam, and Revizto, improving system integration and constructability.
- Collaborated with cross-functional design teams to identify and resolve technical conflicts, maintaining alignment with project specifications and timelines.
- Performed storm load and fixture cost calculations using Excel and Bluebeam, supporting design validation and material estimation.
- Coordinated with partnered firms and clients to address furring and chasing requirements, ensuring adherence to design standards, securing approvals from architectural and construction teams, and supporting cohesive construction planning.

TECHNICAL PROJECTS

Electric Braille Tablet

Mar. 2024

Team Lead - Competition Submission

- Led a multidisciplinary team to develop a Braille tablet capable of reading handwritten input and translating it into Braille using 3D printing, electromagnetic coils, and a machine learning algorithm, enhancing accessibility for visually impaired users.
- Successfully secured both the Top Prize and Audience Choice Award at the SFU Applied Science Speak Out competition, demonstrating innovation and user-focused design.

Automated Home Agriculture Robot

Jan. - Apr. 2024

Mechanical Design Lead

- Designed and modelled a 4-DOF robotic arm on a linear rail capable of autonomous seed planting across scalable garden layouts, improving efficiency in home agriculture automation.
- Developed all components in SolidWorks, optimizing reach and geometry for maximum soil coverage and mechanical reliability.
- Simulated arm motion using MATLAB with Denavit-Hartenberg kinematic modelling, validating performance and motion accuracy before fabrication.

MEMBERSHIPS & CLUBS

Team Phantom Frame and Aero Team - Project Lead

Nov. 2021 - Aug. 2025

- Led manufacturing planning, including using SolidWorks to 3D model the car, designing and creating the welding jig, and general assembly of the project.
- Created and performed tests to ensure accuracy between digital and physical prototyping using various measurement tools and 3D scanning technology.
- Performed 3D printing, laser cutting, and basic welding to manufacture the frame.

Mechatronic Systems Student Society - Vice President

Nov. 2020 - June 2025

- Held leadership roles including VP Academics, VP Services, and VP Professional Relations, providing strategic direction and oversight for student initiatives.
- Served as a liaison between students and faculty, contributing student perspectives to influence institutional decisions.
- Developed and maintained sponsor relationships, securing tens of thousands of dollars to fund student initiatives.
- Organized and represented SFU at large-scale nationwide events, enhancing the university's visibility and student engagement.

CONFERENCES & PROFESSIONAL DEVELOPMENT

SFU Basics of Machine Tools

Aug. 2024

- Completed an intensive 4-day course with hands-on precision machining, using bandsaws, mills, lathes, drills, and metal sheet cutters/benders to produce accurate components.

SFU Agritech BootCamp

Apr. 2024

- Completed a 6-day program focused on robotics and machine learning applications in agriculture.
- Presented work on Machine Learning in Agriculture, demonstrating practical data-driven solutions.

Conference for Sustainable Engineering

Feb. 2021 & Feb. 2024

- Attended lectures and workshops on sustainable engineering practices, including current industry implementations.
- Developed leadership and communication skills through interactive workshops.
- Learned key building standards and requirements as outlined by ASHRAE.

HONOURS & AWARDS

Dean's Honour Roll

Aug. 2025

Recognized for maintaining a TGPA above 3.5.

MSE Equity, Diversity, and Inclusion Champion Award

Sept. 2023 & 2024

Awarded for promoting diversity and inclusion within the community.

SFU Inspiring Women in Applied Science Award

Sept. 2023

Recognized for academic excellence while holding prominent leadership positions.

MSE Leadership Award

Sept. 2023

Awarded for demonstrating strong leadership within the SFU community.

Canada Company Bursary

Aug. 2021

Received for exceptional leadership and teamwork in the National Cadet Advisory Committee.

EDUCATION

BASc: Mechatronic Systems Engineering

Sept. 2020 - Aug. 2025

Simon Fraser University, Surrey - Cooperative Degree

INTERESTS

Camping, Mechanical Keyboard Building, Electric Longboarding, Retro Electronics